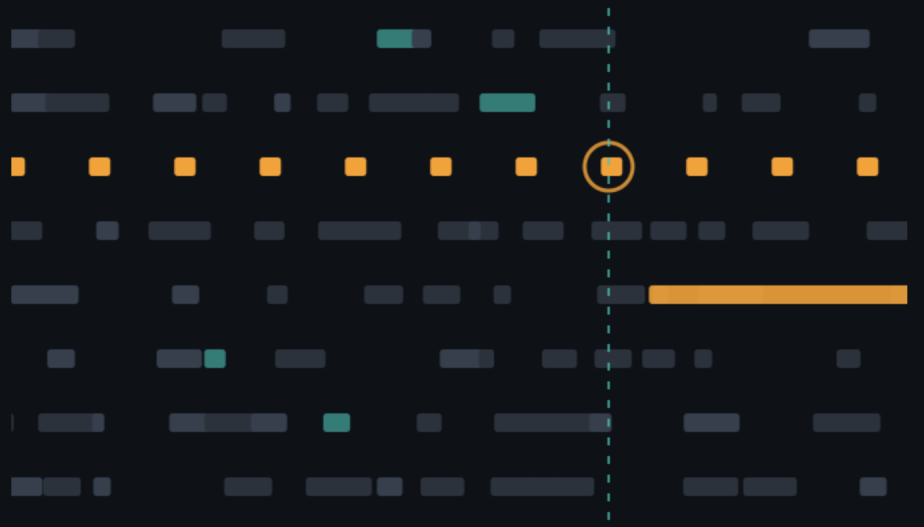
NETWORK TRAFFIC FORENSICS PLATFORM

Net Forensics

Reconstructing the story hidden inside network traffic.



live packet timeline · amber = flagged

01 · PCAP Analysis

Deep packet dissection, L2–L7

02 · Log Correlation

Firewall, server and auth logs on one timeline

03 · Threat Detection

Rules + behaviour + machine learning

Third-year project presentation

01 · THE PROBLEM

Evidence is huge, scattered and quiet

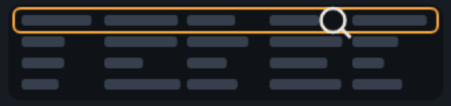
Massive captures

Millions of packets in a single PCAP file.



Manual analysis

Packet-by-packet inspection in tools like Wireshark.



Siloed evidence

Packets, firewall logs and server logs live apart.



Quiet attacks

Beaconing, DNS tunnelling and slow exfiltration blend into normal traffic.



THE RESULT

- Incidents found late
- Patterns missed
- No reproducible evidence trail
- Reports written by hand

THE GAP

No simple tool goes from raw capture to final report in one workflow.

PCAP → sessions → threats → report

02 · THE SOLUTION

Seven modules, one live pipeline



01
Ingest
Upload PCAP and logs; SHA-256 hash for chain of custody

02
Dissect
Decode Ethernet, IP, TCP/UDP, DNS, HTTP, TLS metadata

03
Reconstruct
5-tuple flows, TCP stream rebuild, file extraction

04
Detect
Signatures, heuristics and an anomaly model

05
Correlate
Align logs with flows by time, IP and host

06
Visualise
Timeline, host graph, protocol mix, geo map

07
Report
One-click PDF with evidence hashes

EVIDENCE IN — raw bytes → structured flows

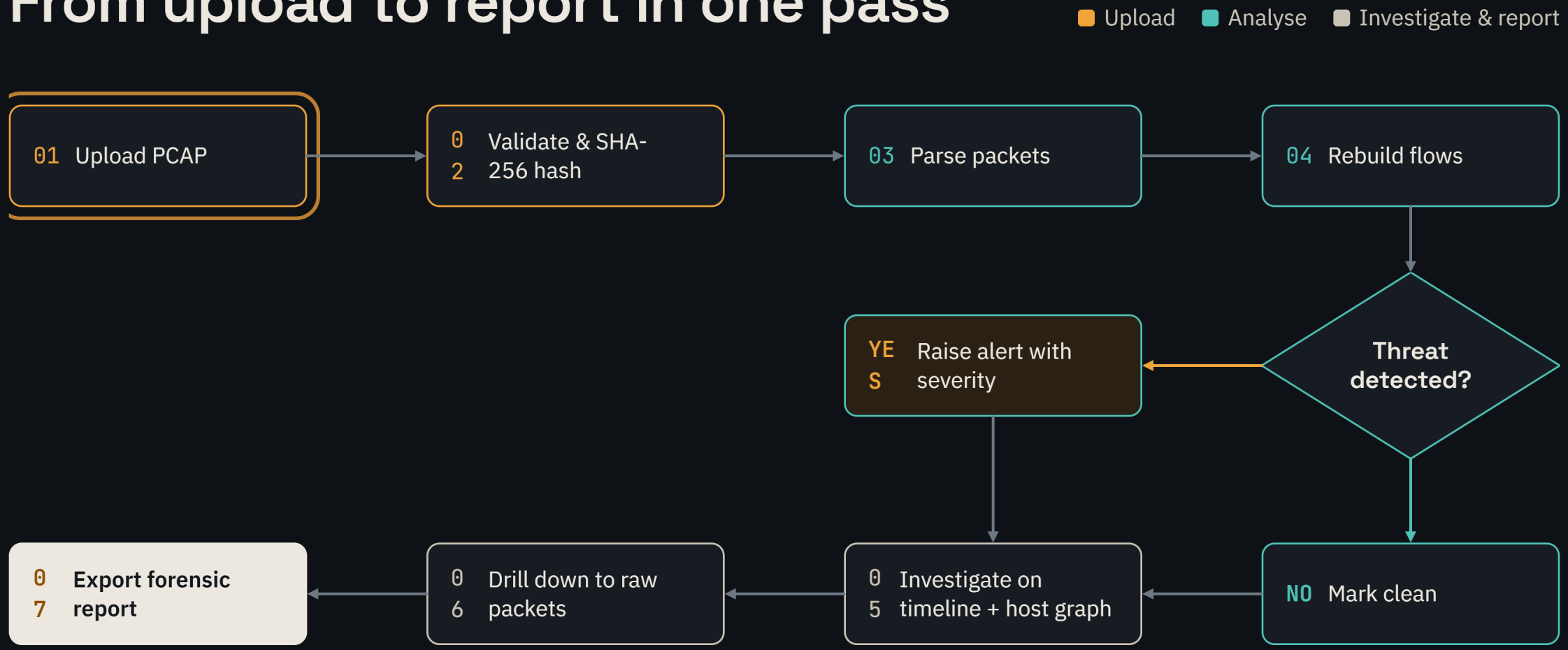
ANALYSIS — flows → threats

OUTPUT — threats → evidence

LIVE · simulated traffic · evidence items 0 · flows 0 · **flagged 0** · all SHA-256 hashed

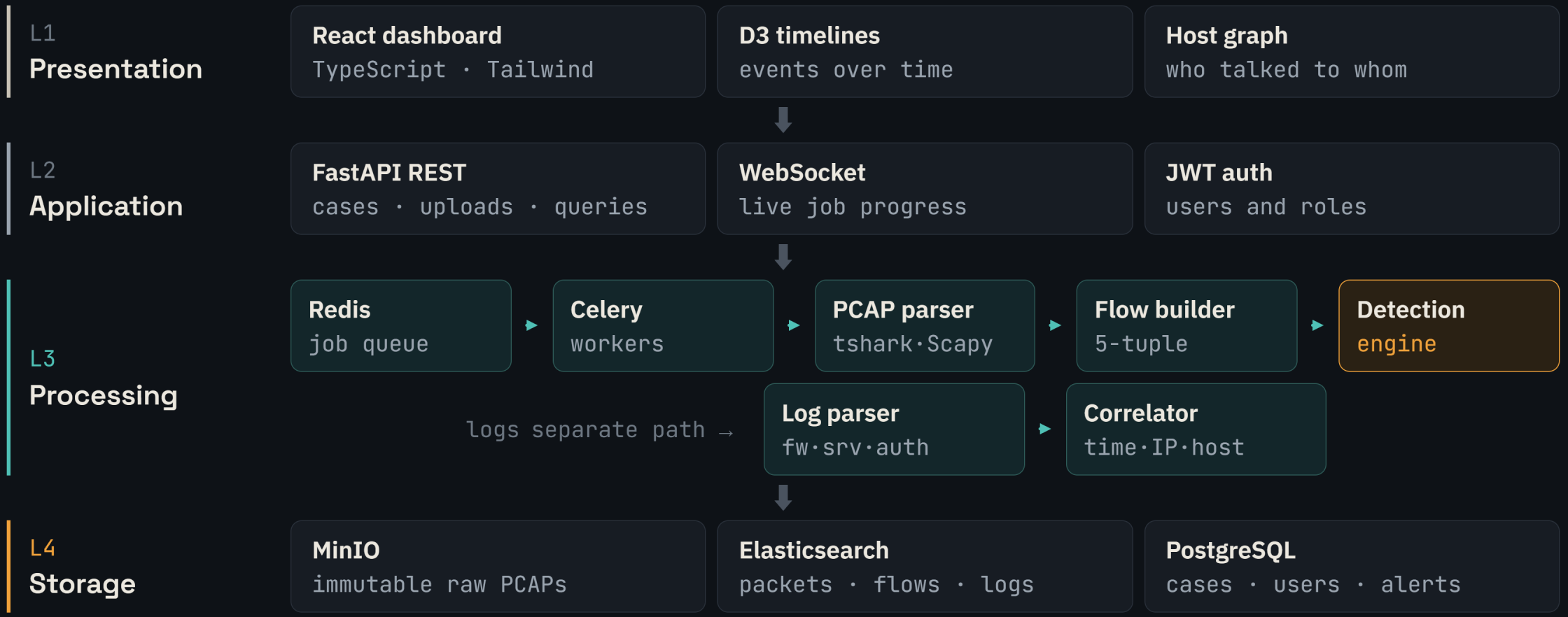
03 · USER FLOW

From upload to report in one pass



04 · SYSTEM ARCHITECTURE

Four layers, fully asynchronous

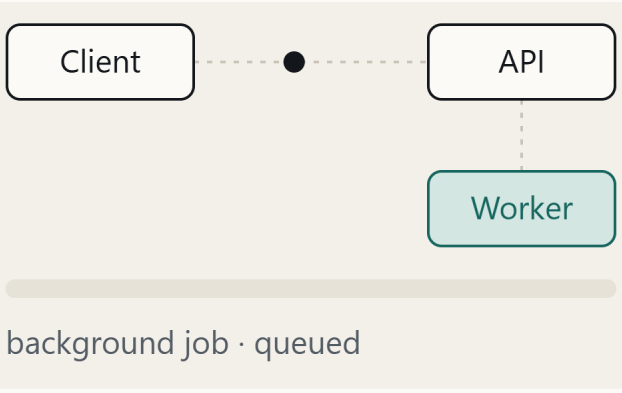


04 · DESIGN DECISIONS

Four choices that keep it fast

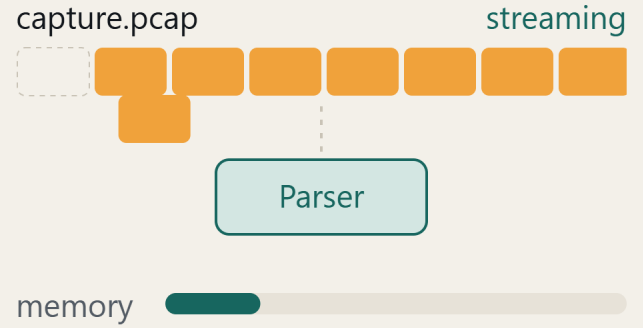
Async pipeline

Uploads return at once; parsing runs behind.



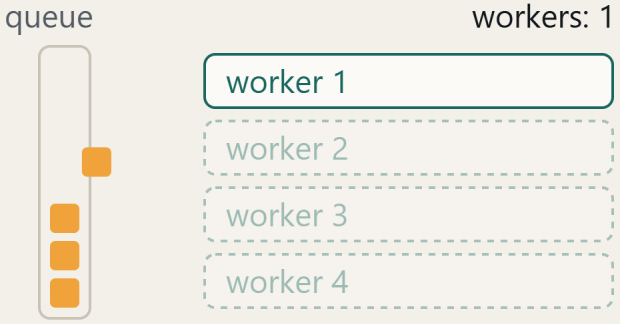
Streamed chunks

Big captures stream in chunks; memory stays flat.



Horizontal workers

More load? Add Celery workers. Nothing else changes.



Fully containerised

One Docker Compose command starts it all.



05 · DETECTION ENGINE

Three layers feed one risk score



05 · WHAT IT CATCHES

Eight threats, each with a measurable signal

Port scan 01

Many ports from one source, unanswered SYNs

`Sliding-window counter`

DNS tunnelling 02

Long high-entropy subdomains, heavy TXT queries

`Shannon entropy`

C2 beaconing 03

Connections at very regular intervals

`Inter-arrival variance`

Data exfiltration 04

Outbound bytes far above baseline

`Z-score deviation`

Brute force 05

Repeated failed logins + matching SSH/RDP flows

`Log-flow correlation`

ARP spoofing 06

An IP suddenly mapped to a new MAC

`State tracking`

Malicious TLS 07

Known-bad client fingerprints

`JA3/JA4 matching`

Unknown anomalies 08

Flows unlike learned normal traffic

`Isolation Forest`

06 · TECH STACK · LIVE REQUEST TRACE

Open-source tools, end to end

PACKET PARSING	BACKEND	JOBS	DETECTION	FRONTEND
tshark, PyShark, Scapy, dpkt	Python, FastAPI	Celery + Redis	Suricata-style rules, scikit-learn, pandas	React, TypeScript, D3.js, Tailwind

SEARCH STORE	DATABASE	EVIDENCE	REPORTS	DEVOPS
Elasticsearch	PostgreSQL	MinIO (S3-compatible)	WeasyPrint (HTML → PDF)	Docker Compose, GitHub Actions

06 · BUILD PLAN

Five phases, tested on public data

PHASE 1

Core parser

Upload PCAP, decode packets, index in Elasticsearch

PHASE 2

Flows + dashboard

Sessions, protocol stats, timeline

PHASE 3

Detection

Rule engine, heuristics, risk scoring

PHASE 4

ML + logs

Isolation Forest, log correlation

PHASE 5

Reports + polish

PDF reports, auth, Docker deployment

TESTED ON

- CIC-IDS2017
- UNSW-NB15
- CTU-13
- Malware-Traffic-Analysis.net

SUCCESS METRICS

- Precision / recall
- Packets per second
- False-positive rate
- Upload-to-report time

07 · CHALLENGES AND FUTURE SCOPE

Hard parts, and where it goes next

Encrypted TLS	Analyse metadata instead: SNI, certificates, JA3/JA4, timing
Huge captures	Streamed parsing and parallel workers
False positives	Combined risk score with tunable thresholds
Evidence integrity	SHA-256 hashing and read-only storage

FUTURE SCOPE

- Live capture with Kafka streaming
- Threat-intel feeds
- Multi-user case management
- Deep learning for encrypted traffic

NET FORENSICS

From raw packets to readable evidence.

Thank you. Questions?

PCAP → flows → threats → report